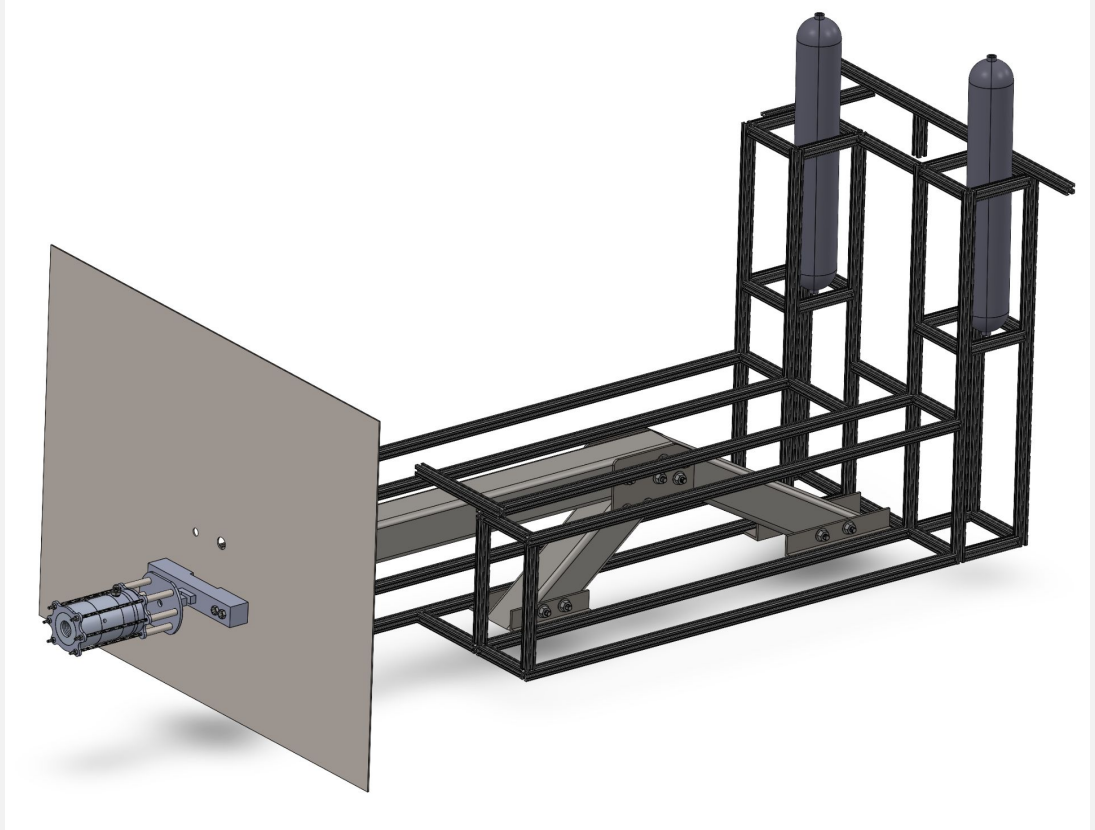


Torito Liquid Propulsion Critical Design Review

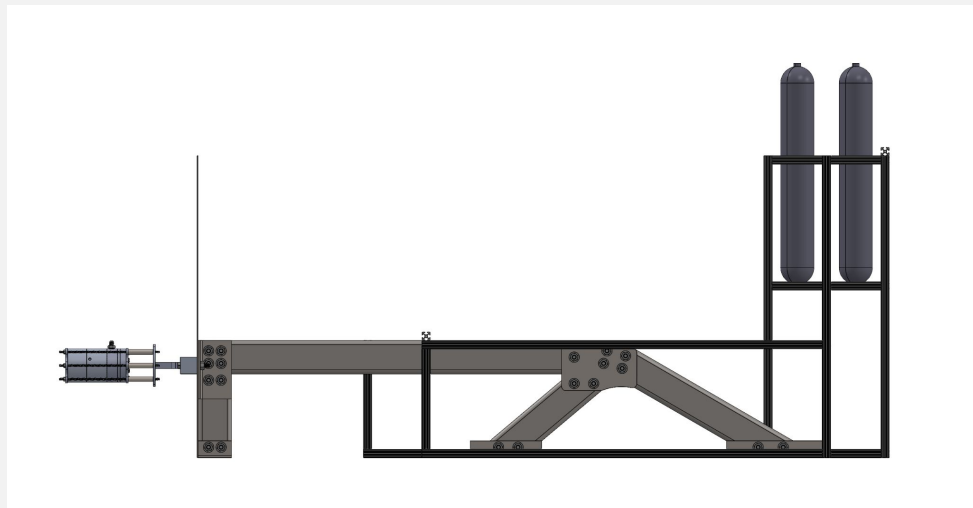
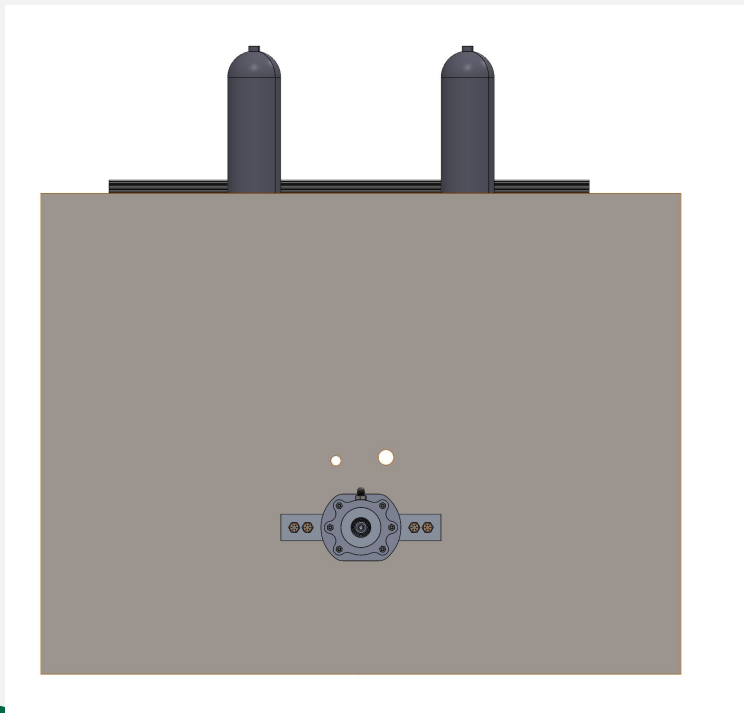
Cesar Briones, Hari Patel, Ethan Zylman, Alston Wang
June 28, 2026

Agenda

1. Introduction
2. Feed System
3. Electrical System
4. Engine
5. Test Stand
6. Safety
7. Water Flow Sequence
8. Logistics



Introduction



Introduction - General Requirements

Req. ID	Requirement	Rationale	Verification Method	Child Req	Parent Req	Status
GEN-1	The engine shall produce at least 500 N of thrust	Setting achievable low stakes goal	Test			In progress
GEN-2	The engine and instruments shall remain undamaged	Protects hardware reusability and ensures collected test data is valid	Analysis and Demonstration	TST-1, TST-2, ENG-1		In progress
GEN-3	The engine and testing instruments shall be able to be transported	Testing location is not USF. Transportation will be required	Demonstration			In progress
GEN-4	The team shall be legally allowed to test	Ensures the test campaign complies with applicable laws	Inspection	LOG-1, LOG-2, LOG-3		In progress
GEN-5	The team shall be able to collect data for post test analysis	Ensures data collection for validation and post test analysis	Test	ECS-1, ECS-2, ECS-4, ECS-6		In progress

Feed System

Feed System - Requirements

Req. ID	Feed System	Rationale	Verification Method	Child Req	Parent Req	Status
FED-1	The feed system shall be able to perform waterflows and cold flows without major changes to the system	Facilitates changes between tests	Demonstration			In progress
FED-2	The Cd and pressure loss of each line shall be known	Makes sure right pressure is being fed into the engine	Analysis, Inspection			In progress
FED-3	The timing delays between pneumatic solenoid actuation and flow confirmation shall be known	Makes sure the oxidizer and fuel enter the chamber at the same time	Inspection		ECS-3	In progress
FED-4	The feed system shall have no leaks	Prevent total system failure, and unexpected pressure losses	Test		GEN-2	In progress
FED-5	The feed system shall have check valves to prevent back flow between lines	Completely separate fuel and ox lines	Inspection		GEN-2	In progress
FED-6	The feed system shall have venting mechanisms on both lines	Remove all fuel and oxidizer in case of emergency stop, or system purge	Inspection		GEN-2	In progress
FED-7	The feed system shall have a passive venting mechanism on the oxidizer side	Prevent overpressurization of ox line	Inspection		FED-12	In progress

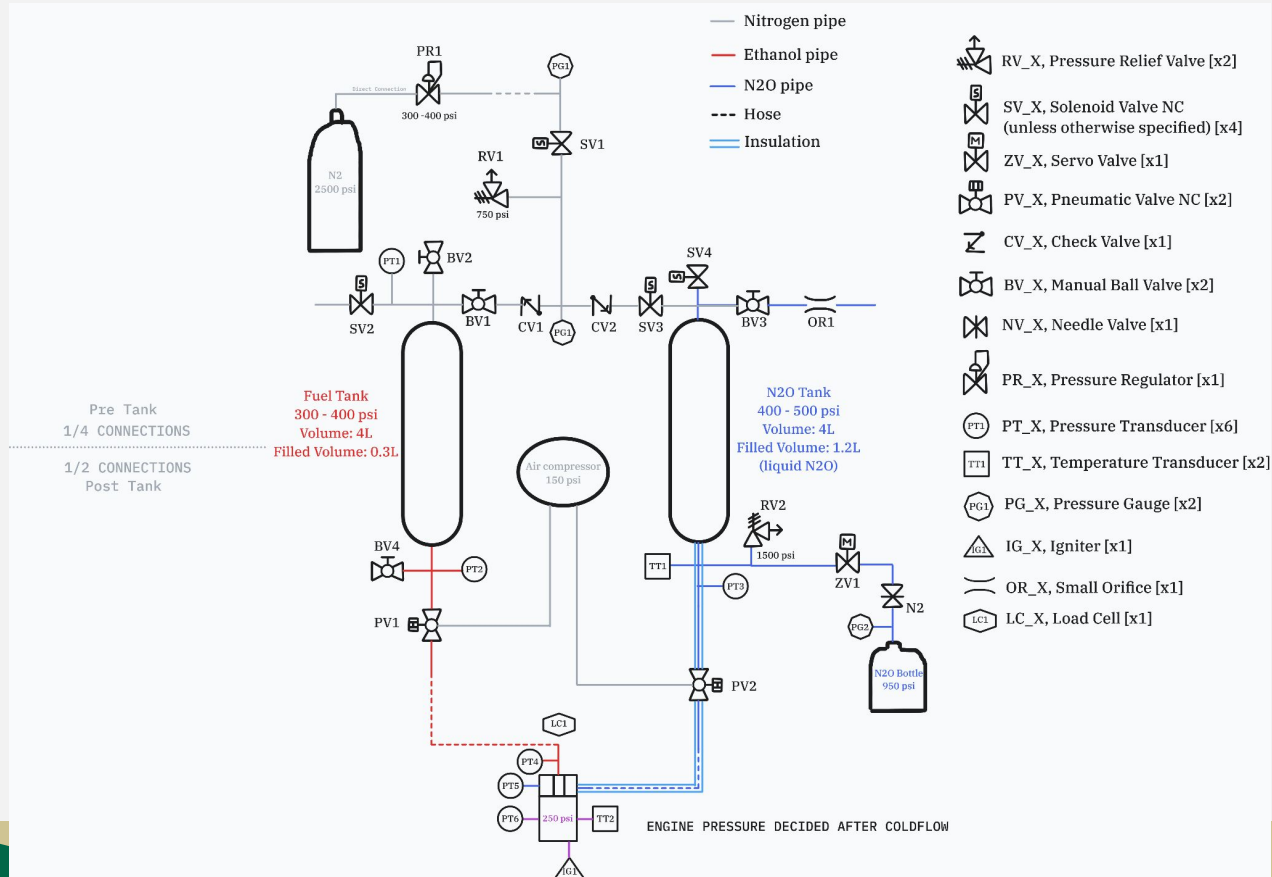
Feed System - Requirements Cont.

FED-8	The oxidizer line shall be insulated	Prevent changes in pressure through the oxidizer line as N2O's pressure is temperature dependent.	Inspection		FED-13	In progress
FED-9	The feed system shall be instrumented with pressure transducers at nitrogen line, run tanks, injector, and chamber	Set locations of pressure collection points	Inspection		ECS-1	In progress
FED-10	The feed system shall be instrumented with temperature transducers at the n2o tank and chamber	Set locations of temperature collection points	Inspection		ECS-2	In progress
FED-11	The feed system shall be capable of remote actuation of all critical valves	No personnel shall be near the feed system while it is being pressurized	Inspection		ECS-5	In progress
FED-12	The feed system shall include relief valves on areas of expected overpressurization	Prevent overpressurization of fuel and ox lines	Inspection	FED-7	GEN-2	In progress
FED-13	The oxidizer line shall maintain liquid N2O delivery to the injector	Injector was designed for liquid N2O	Test, Inspection, Analysis	FED-8	GEN-2	In progress
FED-14	The feed system shall be purgeable	Expell remaining propellant	Test		FED-6	In progress

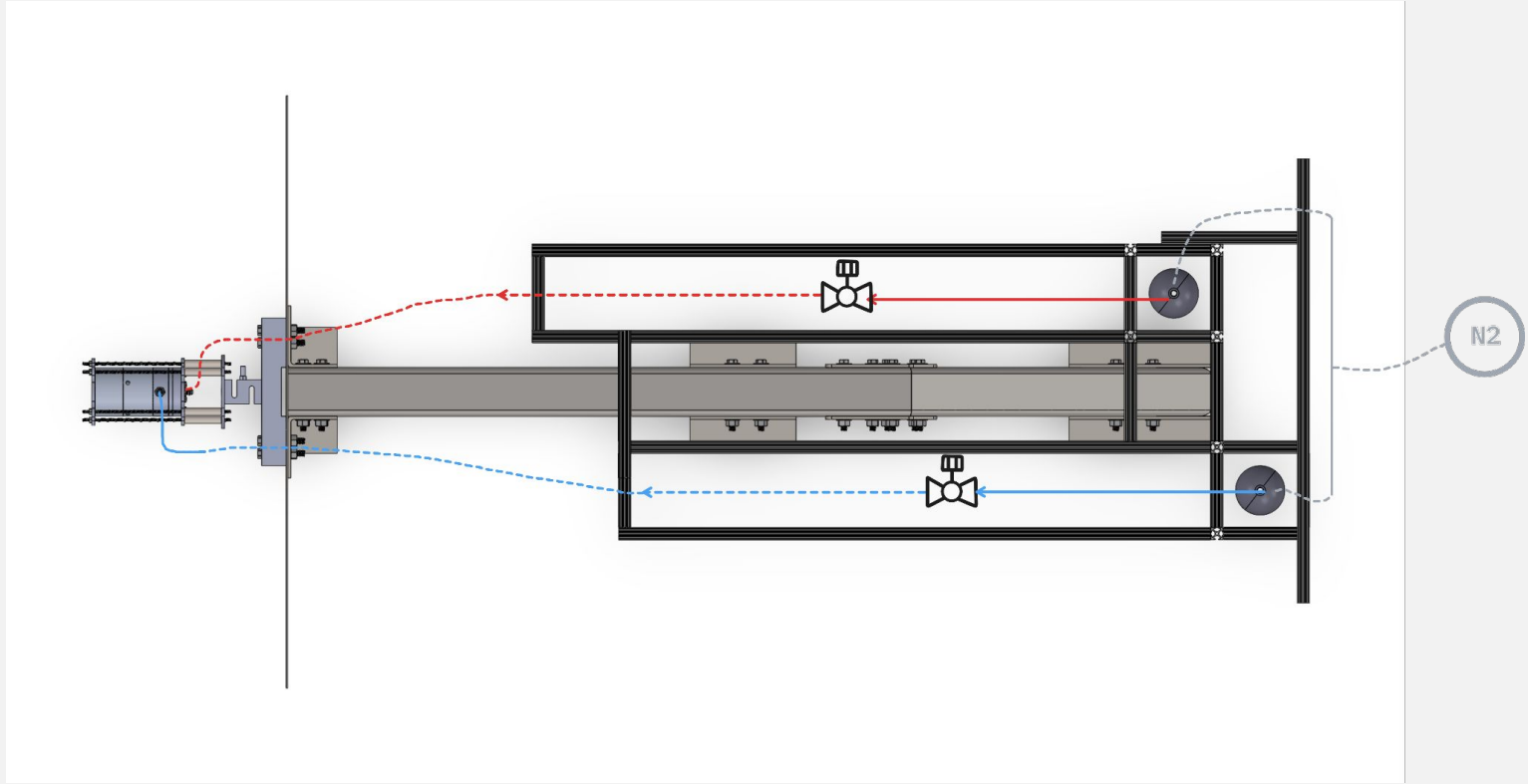
Feed System - Success Criteria

- The feed system delivers fuel to the injector at X pressure with ± 20 psi
- The feed system delivers oxidizer to the injector at X pressure with ± 20 psi
- The feed system delivers propellants to the injector at the same time ± 0.01 s

Feed System - P&ID



Feed System - Piping Location



Feed System - Tanks



Bottles

The system used 2 bottles: one for nitrogen and one for nitrous oxide.



Run Tanks

The team uses run tanks from the old base 11 team. They are 304L Stainless Steel Double Ended DOT-Compliant Sample Cylinders.

Their inner volume is 1gal. Tanks are rated up to 1800 psi.

Feed System - Valves 1



The team uses high pressure solenoids, which are rated up to 5000 psi. McMaster: 1190N24. They have $\frac{1}{4}$ NPT ports on all sides.

Solenoids



The team recently received a sponsorship from Valworx, which provided us with many manual ball valves. They are rated up to 1000 psi.

Run Tanks

Feed System - Valves 2



Check Valves are used to isolate the fuel and oxidizer lines and prevent any backflow. They are rated up to 1000 psi.

Check Valves



The pneumatic valves were put as the final valve before the engine. They were chosen for their high actuation and reliability.

Pneumatic
Valve

Feed System - Control



The team uses 1 regulator after the nitrogen bottle. It is rated for up to 6000 psi inlet and 1200 psi outlet.

Nitrogen Regulator



The team uses swagelok needle valves rated for 5000 psi.

Needle valves

Feed System - Discharge Coefficient

Obtained during water flow test.

Delta time, total mass, pressure, area and density are known.

$$C_d = \frac{\dot{m}_{real}}{\dot{m}_{ideal}}$$

$$\dot{m}_{ideal} = A_2 \sqrt{2\rho(P_1 - P_2)}$$

Electrical System (ECS)

Electrical System - Requirements

Req. ID	Electrical System	Rationale	Verification Method	Child Req	Parent Req	Status
ECS-1	The system shall have 6 pressure transducers capable of reading 0-2000 psig.	Capture pressure readings at a predetermined number of critical points safely.	Inspection	FED-9	GEN-5	In Progress
ECS-2	The system shall have 2 thermocouples capable of reading up to 1000 C.	Capture temperature readings at a predetermined number of critical points safely.	Inspection	FED-10	GEN-5	In Progress
ECS-3	The system shall control 6 solenoids.	Provide remote valve actuation.	Inspection	FED-3	FED-11	In Progress
ECS-4	The system shall store data at a minimum of 10 Hz.	Set minimum Hz for data storage to document events and capture data for post-test analysis.	Test		GEN-5	In Progress

Electrical System - Requirements (cont.)

Req. ID	Electrical System	Rationale	Verification Method	Child Req	Parent Req	Status
ECS-5	The system shall be able to be operated remotely at a distance of at least 500 ft.	Meet compliance for clearance zones.	Test	FED-11	GEN-4	In Progress
ECS-6	The system shall be able to transmit data at a minimum of 100 Hz.	Set minimum Hz for telemetry to confirm	Test		GEN-5	In Progress
ECS-7	The delay between command and actuation should be no more than 25 ms.	Support real-time actuation for precise control & Emergency Stop events.	Test			In Progress

Electrical System - Requirements (cont.)

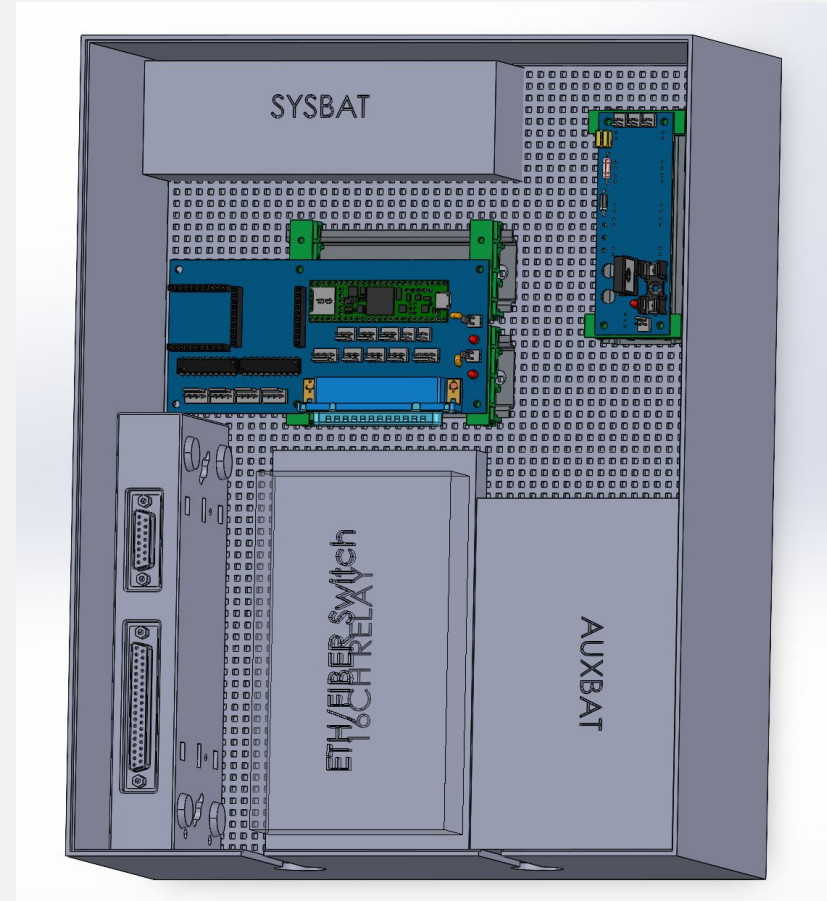
Req. ID	Electrical System	Rationale	Verification Method	Child Req	Parent Req	Status
ECS-8	The system shall support pre-configured actuation sequences with 10 ms precision.	Support automatic actuation for reliable & repeatable results.	Test			In Progress
ECS-9	The system shall Emergency Stop in the case of test-specific predetermined events.	Provide a fail-safe mechanism to meet safety requirements.	Test			
ECS-10	The system shall have a load cell capable of reading up to 3 kN.	Capture thrust readings safely.	Inspection		GEN-1	

ECS - Success Criteria

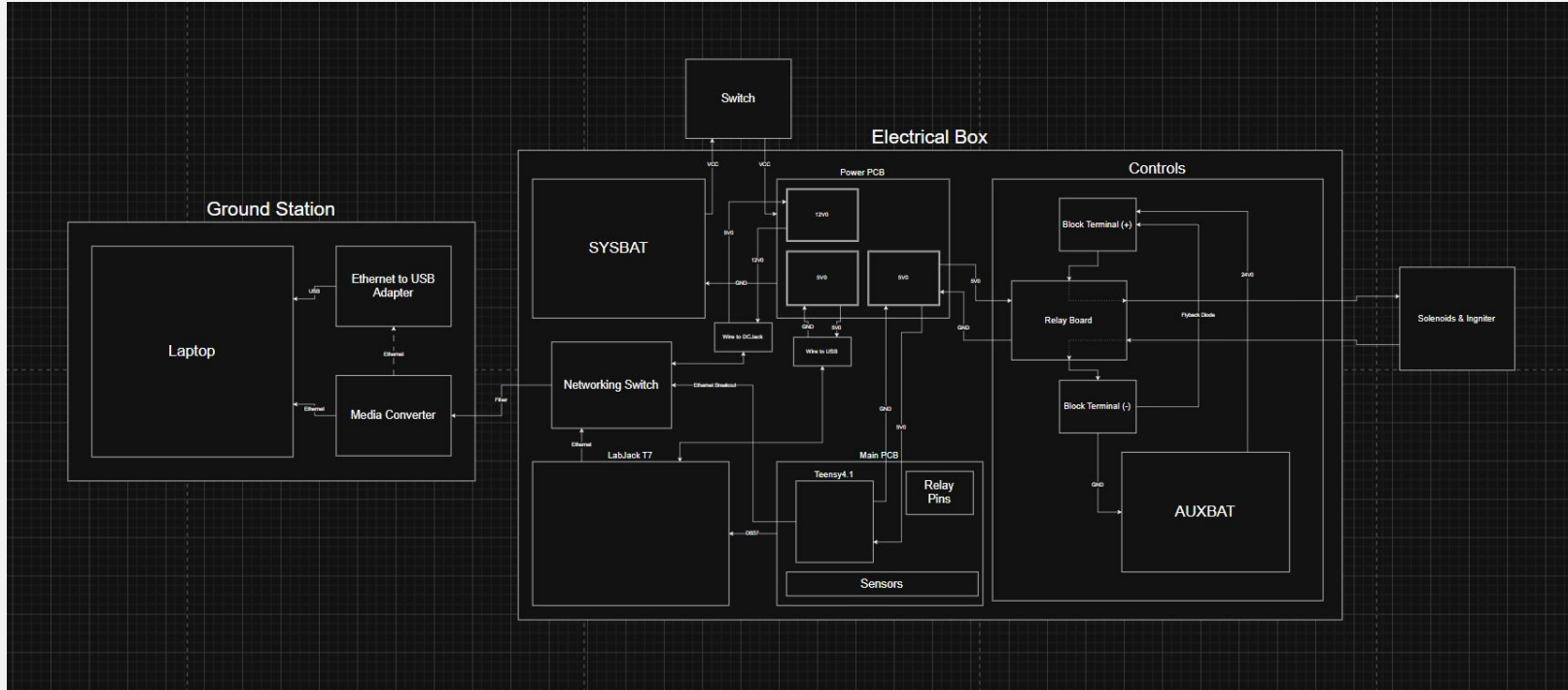
- The system supports remote operation up to 800 ft.
- The delay between command and actuation is no more than 10ms.
- Pre-configured actuation sequences have 5ms precision.
- All three transmission protocols are functional.
- Data transmits at a minimum of 150Hz and stores at 20 Hz.

ECS - Overall Design

- Contained within single, mountable box.
- Two PCB's (*Power & Main*)
- LabJack T7 (DAQ Toolkit)
- Future-Proofing (16 Relay & 14 AI Channels; MUX)



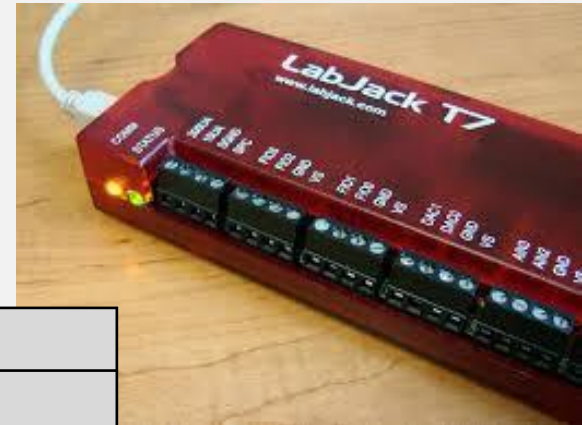
ECS - System Architecture



ECS - Sensing

The system uses the LabJack T7 to collect pressure, temperature, and force data. These measurements provide live feed and post-test data for validating engine behavior.

- Pressure Transducers
- Thermocouples
- Load Cell



Specification	Value
Analog Inputs	14
Analog Resolution	16-bit
Max Stream Rate	100 kSamples/s Aggregate

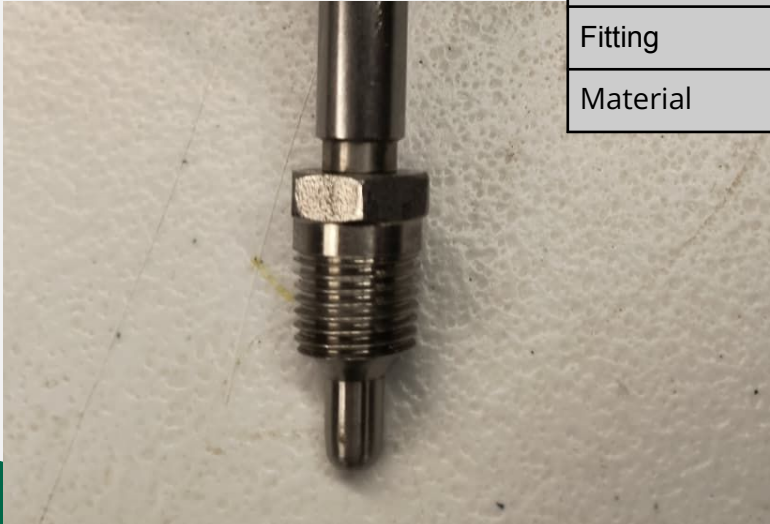
ECS - Pressure Transducers

Specification	Value
Pressure Range	0-2000 psig
Voltage Range	0.5-4.5V
Accuracy	±0.5% Full Scale
Fitting	1/8 NPT
Material	316 Stainless Steel



ECS - Thermocouples

Specification	Value
Type	T, Single Element
Pressure Rating	2500 psi
Temperature Range	0-370 C
Fitting	1/4 NPT
Material	316 Stainless Steel



ECS - Load Cell

Specification	Value
Type	S, 4-Lead
Force Rating	300 kgf



ECS - Transmission Modes

The system uses three transmission modes for remote operation:

- Ethernet/Fiber
- XBee Pro S3B
- RFM9x LoRa

Ethernet/Fiber is the primary high-rate link, while XBee and RFM9x provide RF fallbacks.

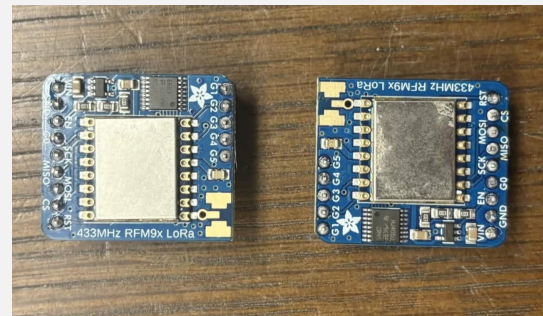
ECS - Fiber Optic/Ethernet

The fiber link carries Ethernet between the ground station and remote switch. This provides the primary, low-latency path for LabJack data and Teensy commands.



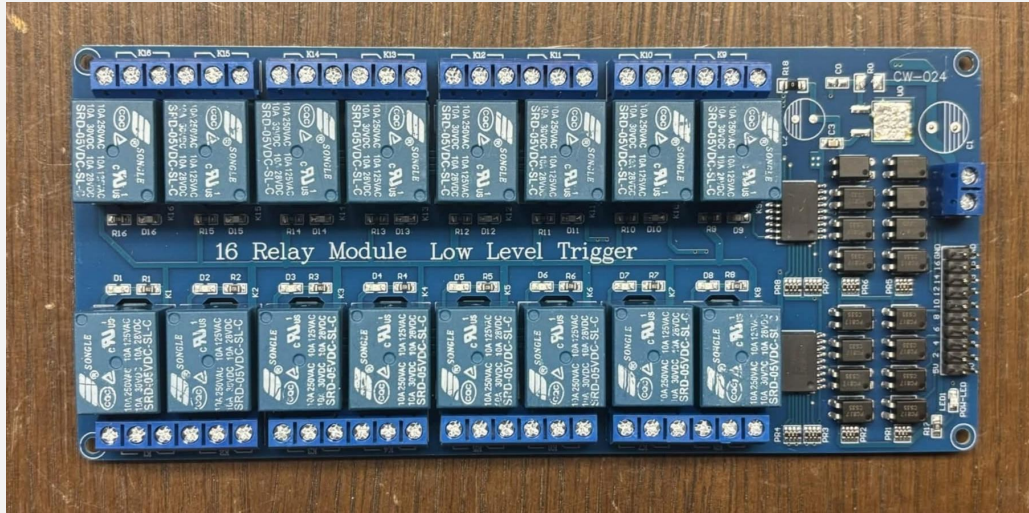
ECS - RF Modules

- XBee provides the primary RF backup. It supports commands, and reduced-rate telemetry if the fiber link fails.
- RFM9x provides the final low-bandwidth fallback. It prioritizes commands over telemetry.



ECS - Controls

The Teensy 4.1 receives commands and controls the relay board. The relays switch the solenoid valve circuits, allowing remote valve actuation, pre-configured sequences, and Emergency Stop control.

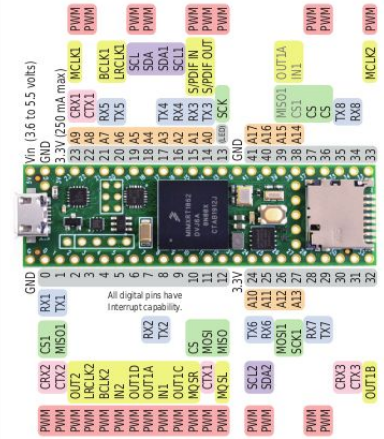


Welcome to Teensy® 4.1

32 Bit Arduino-Compatible Microcontroller

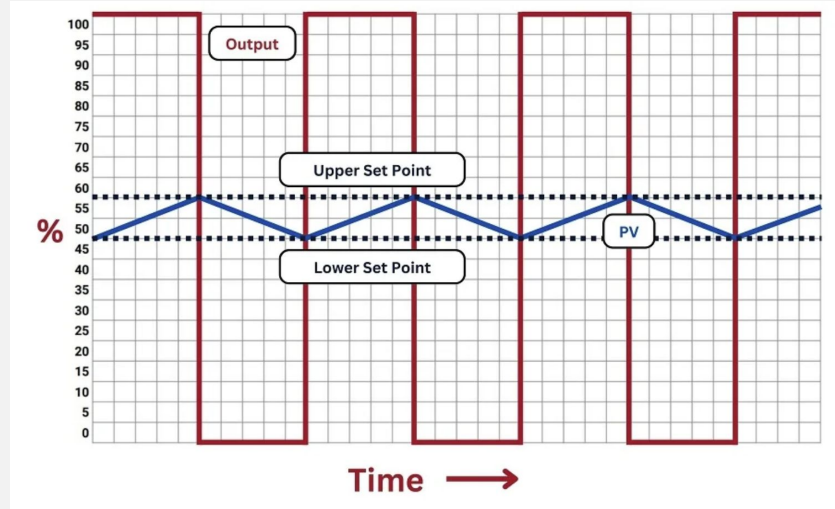
To begin using Teensy, please visit the website & click [Getting Started](#).

www.pjrc.com/teensy



ECS - Bang Bang Mechanism

The bang-bang mechanism regulates N₂O run tank pressure using a solenoid vent valve. When pressure rises above the upper limit, the valve opens. When pressure falls below the lower limit, the valve closes.



ECS - Emergency Stop & Ignition

The Emergency Stop sequence moves the system to a predetermined safe valve state if command integrity is lost. Main valves close while selected relief valves open to reduce system pressure.

Ignition is controlled through a relay-driven circuit. The circuit activates a COTS e-match surrounded by solid propellant grains to begin ignition.



Engine

Engine - Requirements

Req. ID	Requirement	Rationale	Verification Method	Child Req	Parent Req	Status
ENG-1	The engine shall maintain structural integrity when exposed to combustion gas temperatures up to 3000 K for the full duration of the static fire test.	Chamber will be exposed to hot gasses during burning	Analysis		GEN-2	In progress
ENG-2	Most of the engine components shall be manufactured at the shop	Reduce costs and lead time for manufacturing	Demonstration			In progress
ENG-3	The engine shall use Nitrous Oxide as oxidizer	Already available oxidizer	Inspection			Complete
ENG-4	The engine shall use Ethanol as fuel	Already available fuel	Inspection			Complete

Engine - Success Criteria

- The engine is able to sustain thrust for 3 seconds
- No visible deformation, cracking, or burn-through of the combustion chamber or nozzle shall occur during or after the test
- All seals and O-rings shall remain functional

Engine - Performance

Engine performance is highly dependent on pressure. The engine was designed for a 12 bar inlet. However, the team realized too late that receiving liquid nitrous oxide at 12 bar for the injector would be unrealistic. The engine walls, however, are thick enough to sustain much higher pressures.

Chamber pressure is expected to be in between 15-40 bar.

Engine - Performance Cont.1

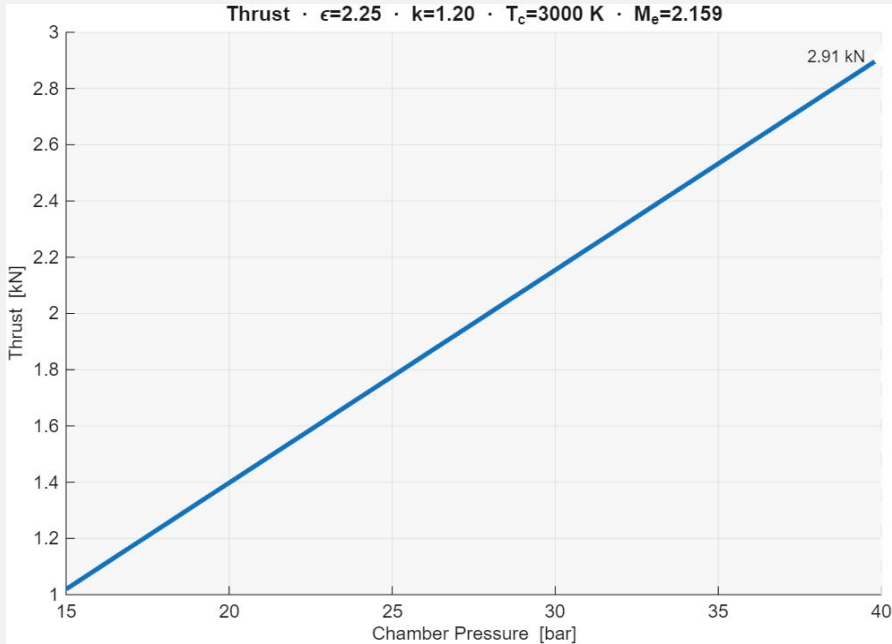
Using the following formulas the team was able to sweep for thrust, mass flowrate and total propellant along this range.

$$\dot{m} = \frac{P_1 A_t}{\sqrt{T_1}} \sqrt{\frac{k}{R}} \left(\frac{2}{k+1} \right)^{\frac{k+1}{2(k+1)}}$$
$$F = \dot{m} V_2 + (P_2 - P_{amb}) A_2$$

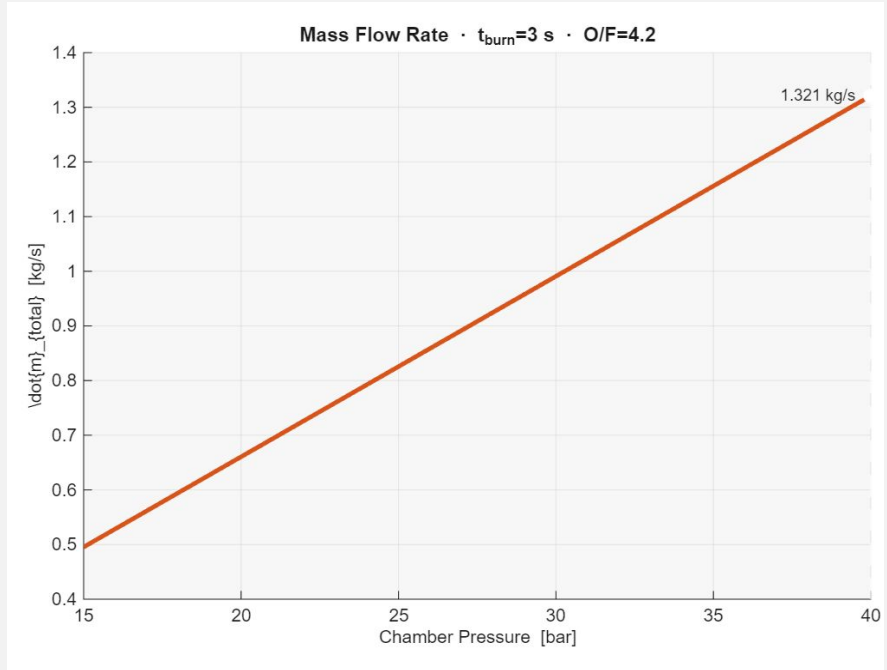
Engine - Performance Cont.2

Parameter	Symbol	Value	Units
Throat radius	$r_\square$	0.0127	m
Exit radius	r_2	0.01905	m
Throat area	$A_\square$	5.067×10^{-4}	m^2
Exit area	A_2	1.140×10^{-3}	m^2
Area ratio	ϵ	2.25	—
Chamber temperature	T_1	3000	K
Specific heat ratio	k	1.2	—
Specific gas constant	R	330	J/kg·K
Ambient pressure	P_{amb}	101,325	Pa
O/F ratio	O/F	4.2	—
N ₂ O liquid density	ρ_{ox}	900	kg/m ³
Ethanol liquid density	ρ_{fuel}	789	kg/m ³
Burn duration	t_{burn}	3	s

Engine - Performance Cont.3

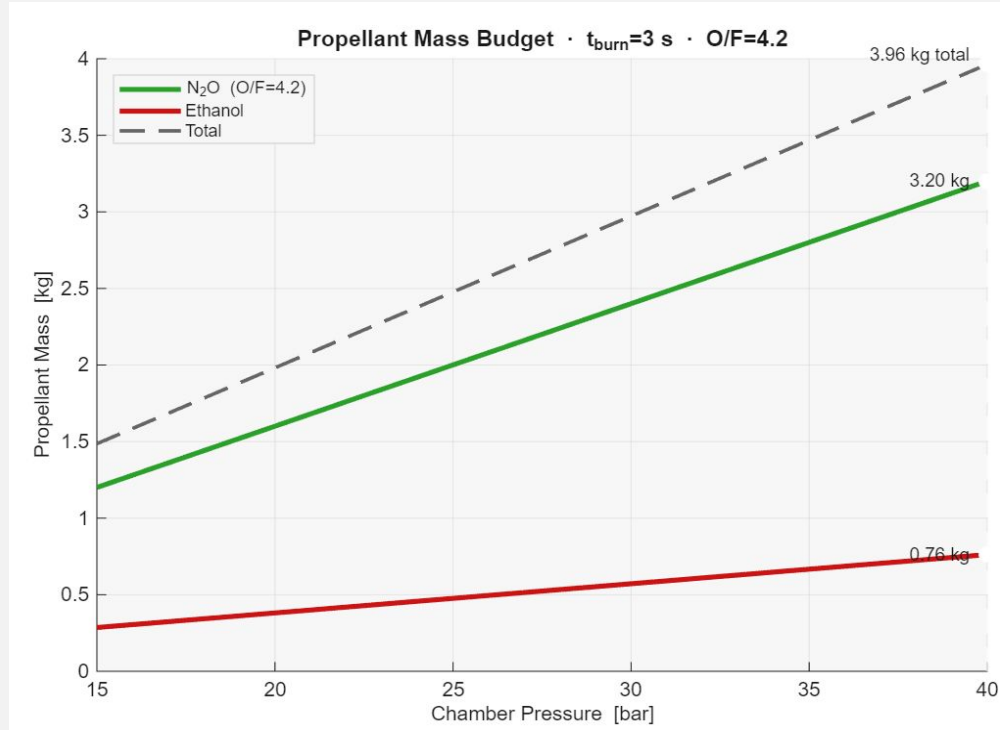


Force vs Chamber Pressure



Mass Flow rate vs Chamber Pressure

Engine - Performance Cont.4



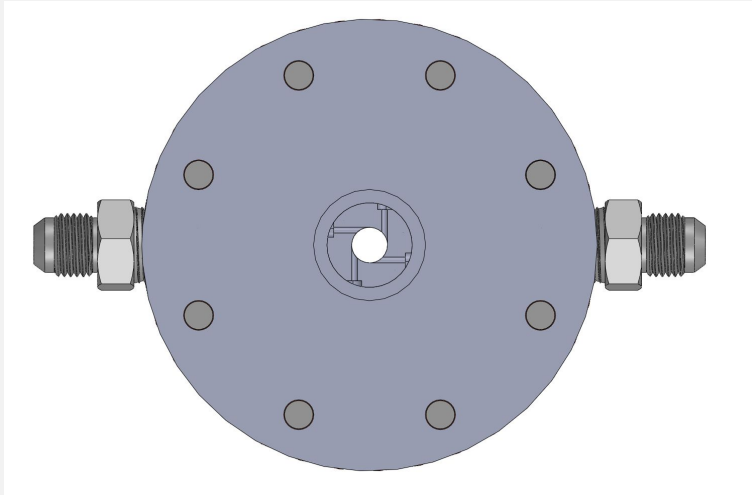
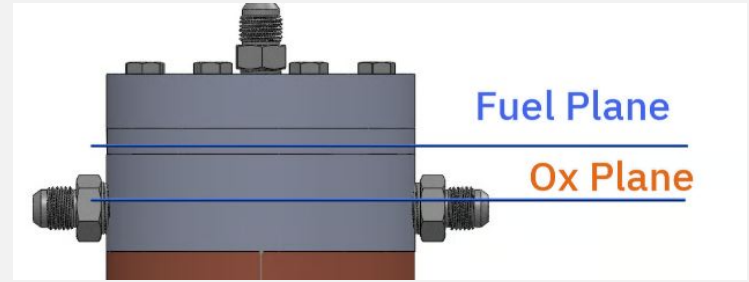
Propellant mass vs Chamber Pressure

Engine - Injector

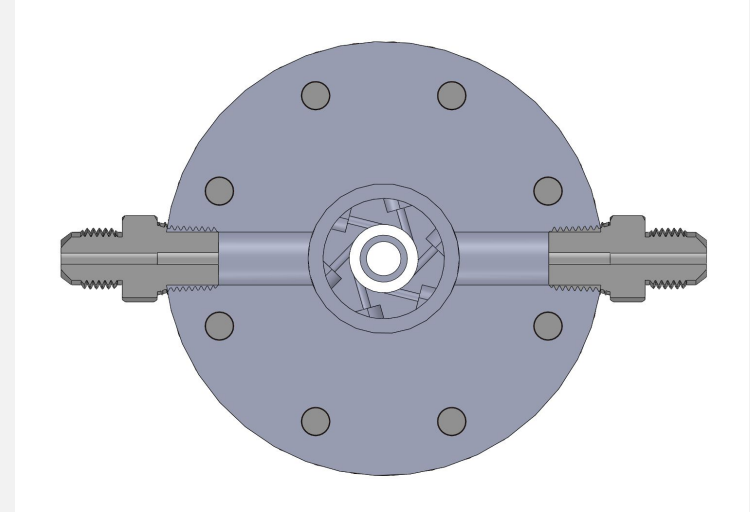
Coaxial Swirl Injector single element. Manufactured by JLCCNC



Engine - Injector Cont.2

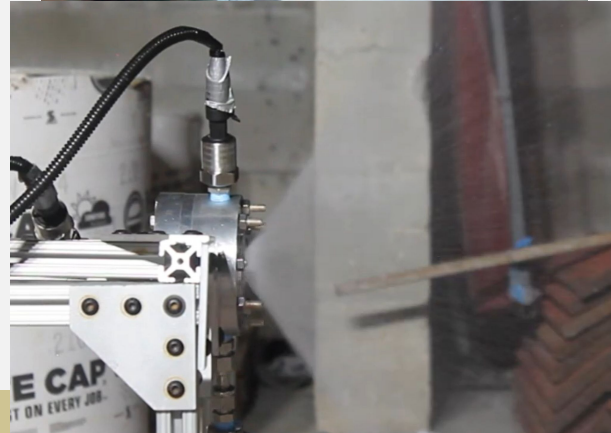
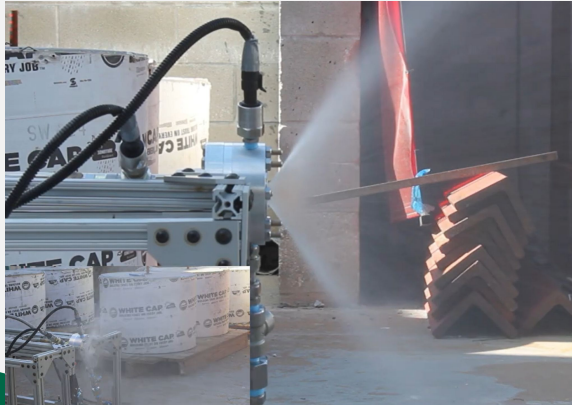
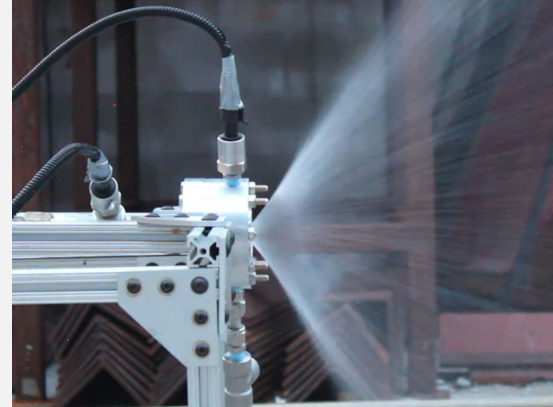


Fuel Plane



Oxidizer Plane

Engine - Injector Testing



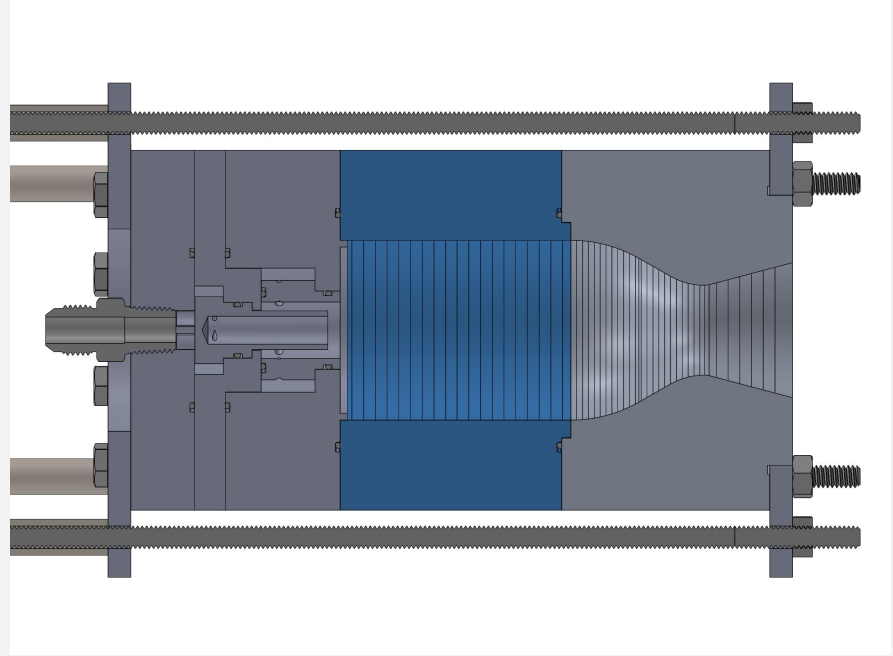
Engine - Injector testing Cont.



Engine - Injector Testing Cont.2

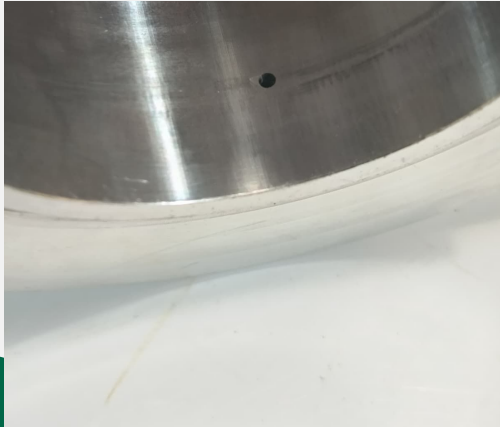
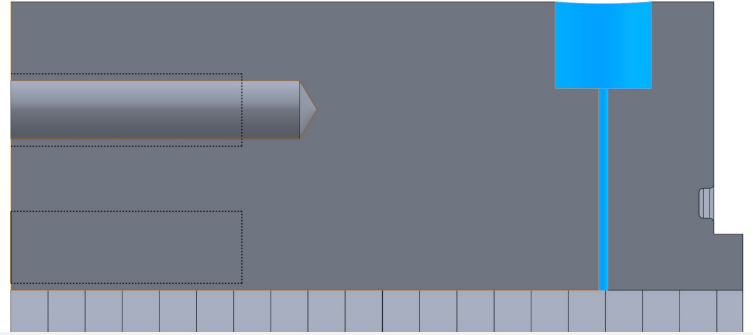
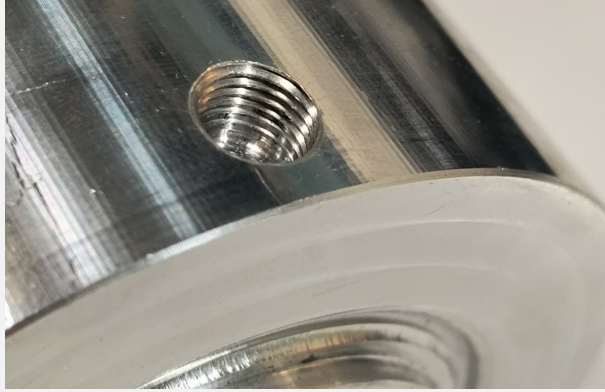


Engine - Chamber

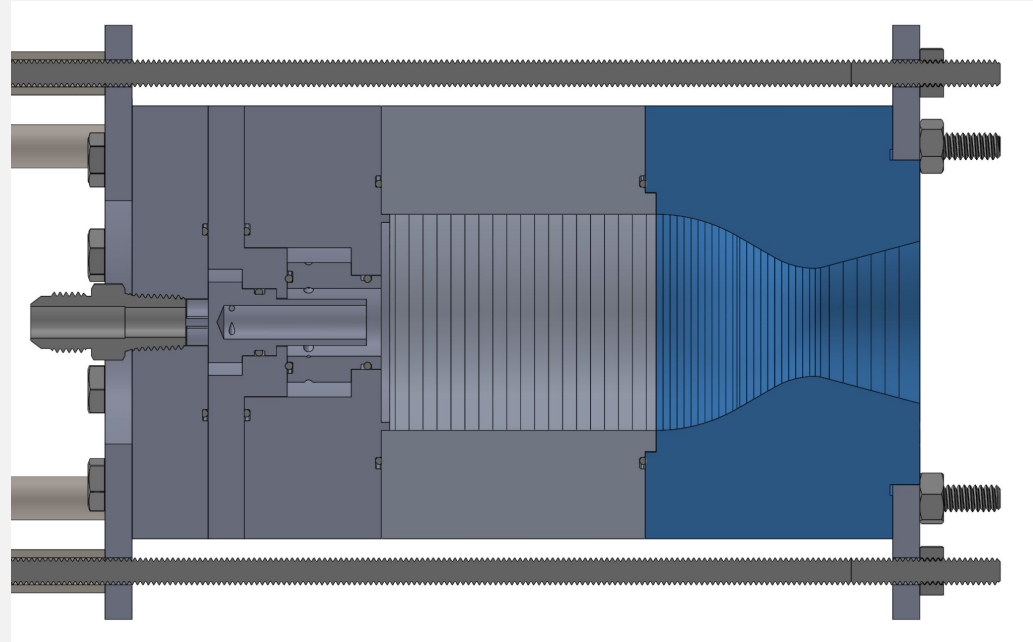


Engine - Chamber Cont.1

0.45 in depth
for full
engagement



Engine - Nozzle



Engine - Diverging Nozzle Section

In order to know the exit diameter of the nozzle, we need to know the exit mach velocity.

$$p_0 = p[1 + \frac{1}{2}(k - 1)M^2]^{k/(k-1)}$$

$$900000 = 91192.5[1 + \frac{1}{2}(1.157 - 1)M^2]^{1.157/(1.157-1)}$$

$$M = 2.154$$

$k = 1.157 = C_p/C_v$ of mix propellant

Engine - Diverging Nozzle Section Cont.1

The relationship between the mach number and area is known through the following equation.

$$\frac{A_y}{A_x} = \frac{M_x}{M_y} \sqrt{\left\{ \frac{1 + [(k-1)/2] M_y^2}{1 + [(k-1)/2] M_x^2} \right\}^{(k+1)/(k-1)}}$$

Y = @nozzle exit
X = @nozzle throat

$$\frac{A_2}{A_t} = \frac{1}{2.154} \sqrt{\left\{ \frac{1 + [(1.157-1)/2] 2.154_y^2}{1 + [(1.157-1)/2] 1_x^2} \right\}^{(1.157+1)/(1.157-1)}}$$

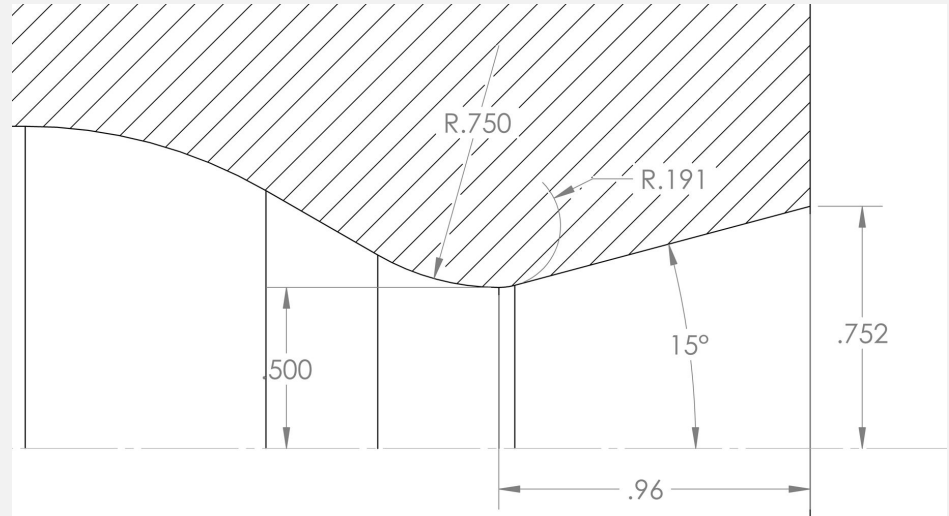
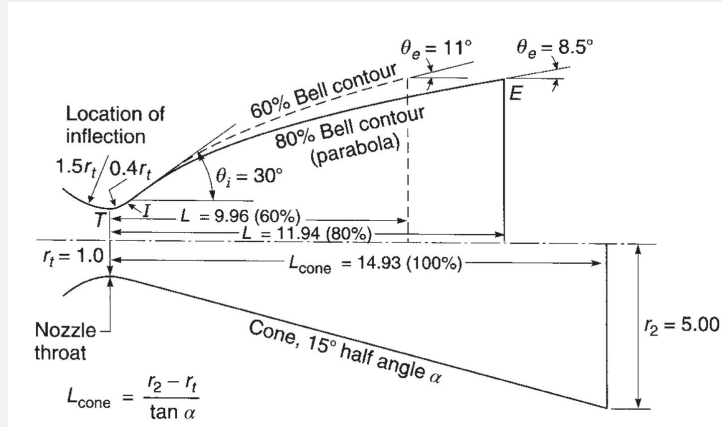
$$\frac{A_2}{A_t} = 2.33, r_2 = 0.752 \text{ in}$$

Velocity at throat is Mach 1!!

k = 1.157 = Cp/Cv of mix propellant

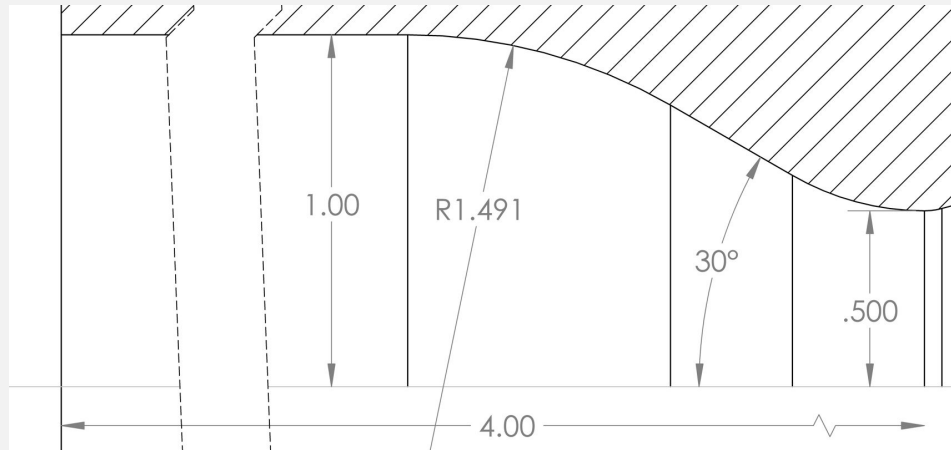
Engine - Diverging Nozzle Section Cont.2

By following the cone, 15° half angle, the following nozzle can be constructed



Engine - Diverging Nozzle Section Cont.3

As described in RPE, the converging nozzle section shape does not significantly affect nozzle performance. "Its subsonic flow can easily be turned with very low pressure drops and almost any given radius, cone angle, wall contour curve, or nozzle inlet shape satisfactory."



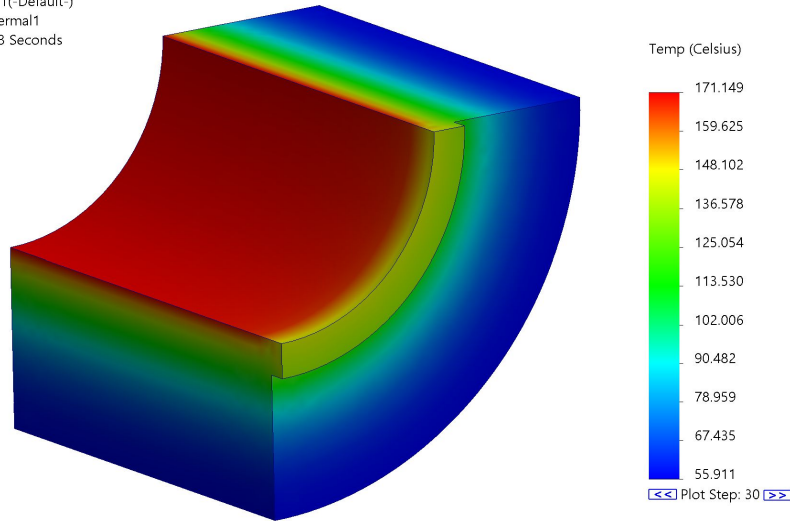
Engine - Thermal Analysis

Gas temp: 3000K

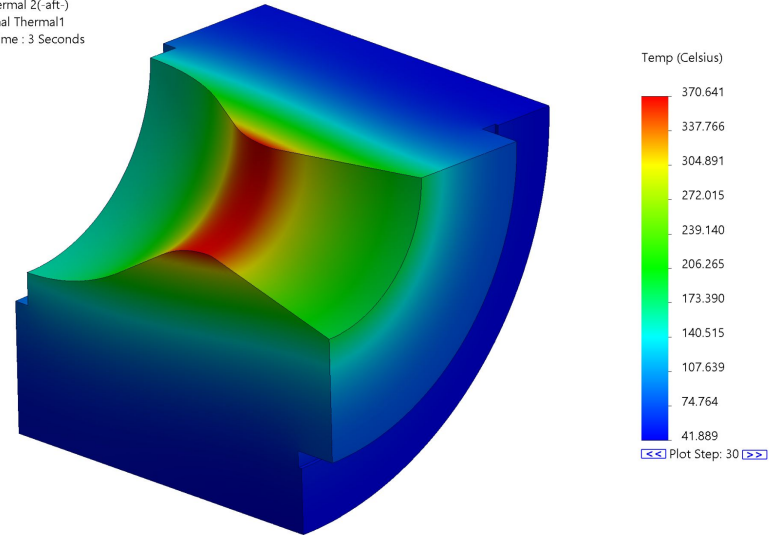
Convection heat transfer coefficient: 700 to 2100 W/m²K

Duration: 3 seconds

Model name: Combustion Chamber simd
Study name: Thermal 1(-Default-)
Plot type: Thermal Thermal1
Time step: 30 time : 3 Seconds

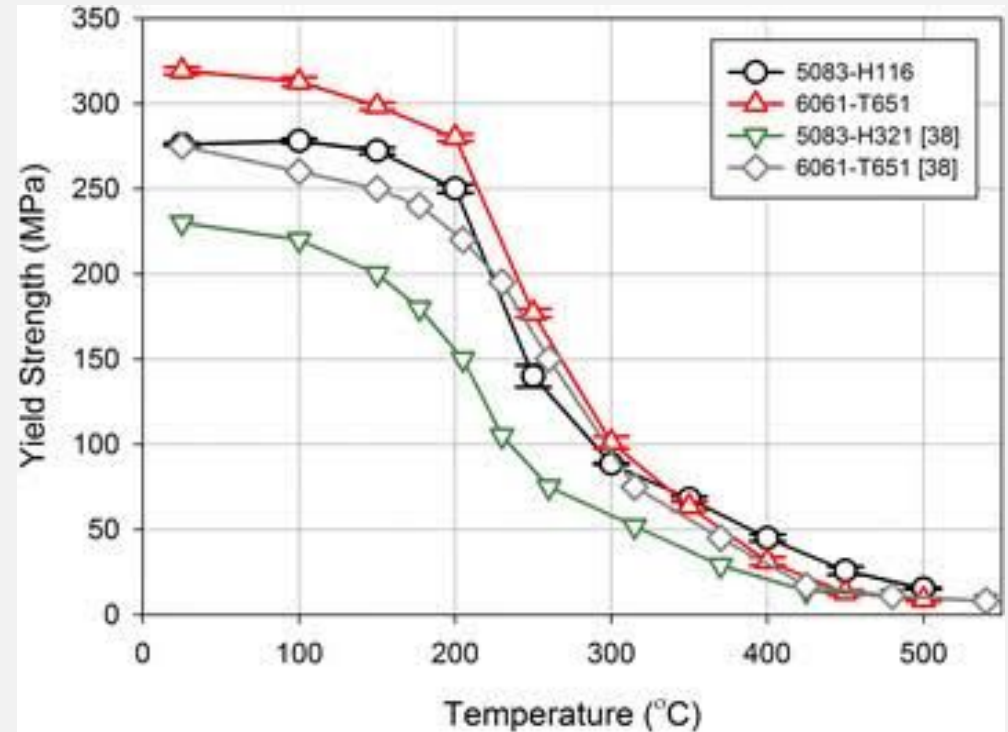


Model name: Combustion Chamber simd
Study name: Thermal 2(-aft-)
Plot type: Thermal Thermal1
Time step: 30 time : 3 Seconds



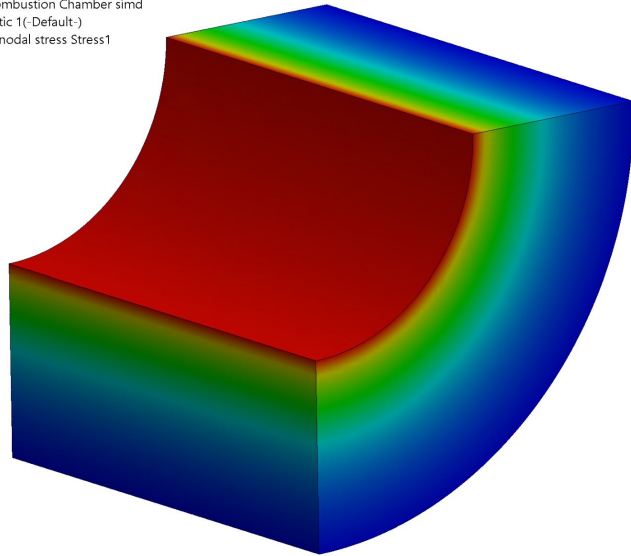
Engine - Structural Analysis

Material's yield stress decrease as temperature increases.

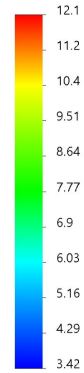


Engine - Structural Analysis

Model name: Combustion Chamber simd
Study name: Static 1(- Default-)
Plot type: Static nodal stress Stress1

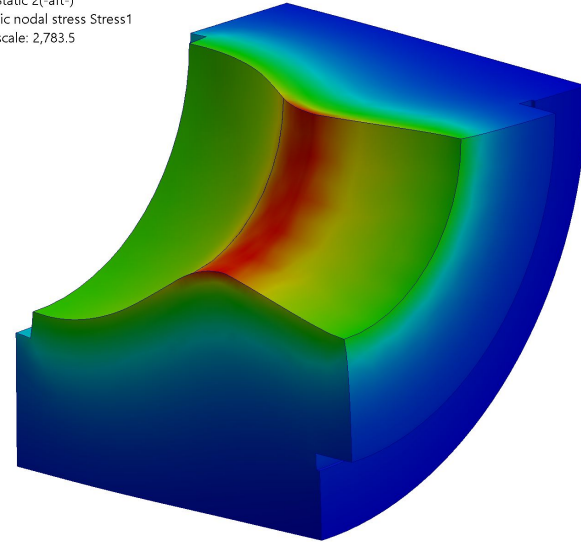


von Mises (N/mm² (MPa))

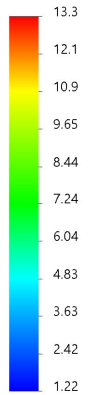


→ Yield strength: 275

Model name: Combustion Chamber simd
Study name: Static 2(- aft-)
Plot type: Static nodal stress Stress1
Deformation scale: 2,783.5



von Mises (N/mm² (MPa))



→ Yield strength: 275

3.15 FOS at 370°C at 750 psi

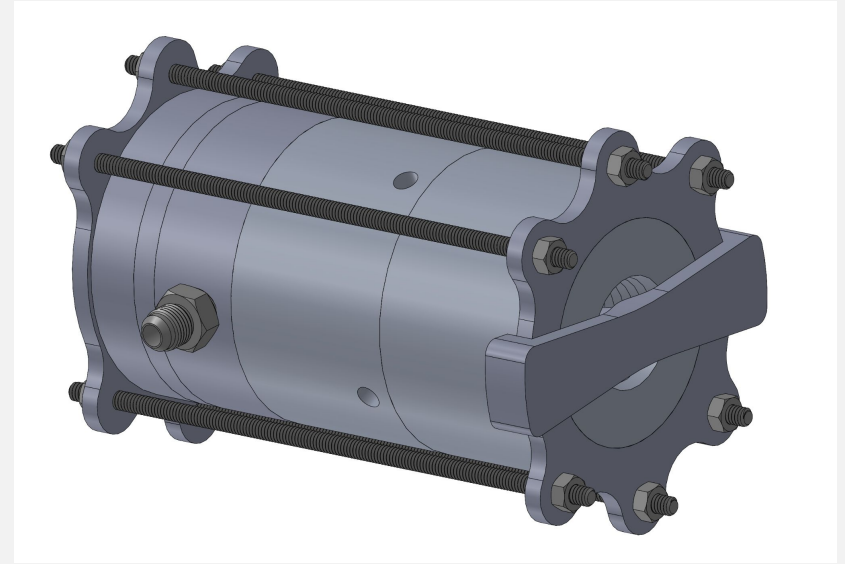
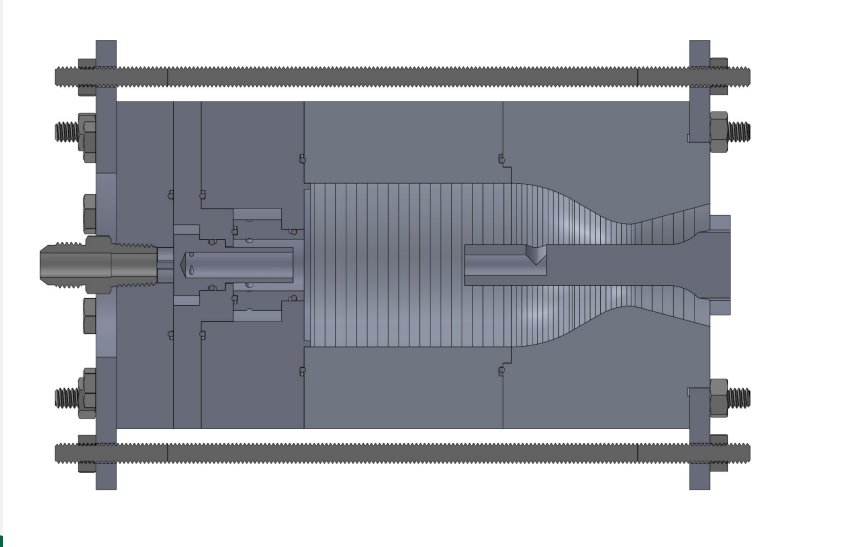
Engine - Manufacturing

Thank you Riccardo!



Engine - Igniter Holder

PLA 3D print that will hold the igniter in the center of the chamber.



Test Stand

Test Stand - Requirements

Req. ID	Requirement	Rationale	Verification Method	Child Req	Parent Req	Status
TST-1	Test stand shall provide support for all other subsystems	All subsystems will be attached to the test stand	Demonstration	TST-1.1, TST-1.2	GEN-2	In progress
TST-1.1	The test stand shall provide support for the run tanks	Run tanks will be located in the test stand	Demonstration		TST-1	In progress
TST-1.2	The test stand shall provide support for the feed system	Feed system will be located in the test stand	Demonstration		TST-1	In progress
TST-2	All structural parts shall have a minimum FS of 2	Avoid test stand mechanical failure	Analysis		GEN-2	In progress
TST-3	The test stand shall be fabricated with available material	Maximize reusability of old components	Inspection			In progress

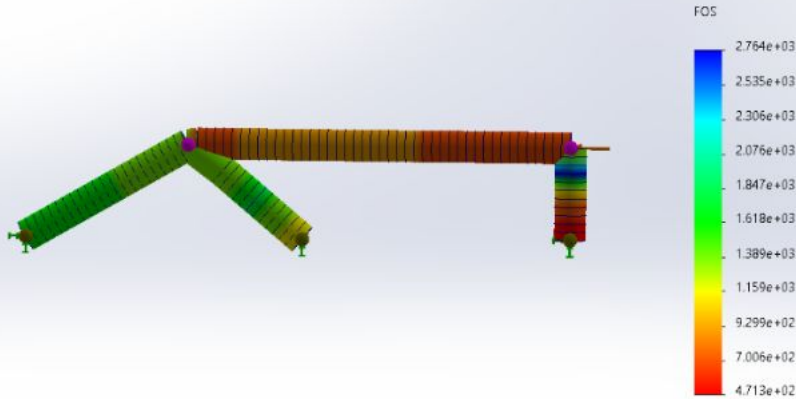
Test Stand - Design Heritage

The truss of the test stand was designed and manufactured by Chris Lemonds two years ago. Even though he is no longer in the team, his contributions will make this test possible.



Test Stand - Design Heritage Cont.1

Factor of safety distribution: Min FOS = 4.7e+02



Structural analysis was done on the assumption of a 250 lbf engine thrust. The maximum loading scenario for the Epimetheus engine is 675 lbf. Assuming a linear and elastic stress, the obtained factor of safeties can be divided by 2.7.

Relative Outputs Presentation		
--- Bolt Related Stress ---		
Output	Value	Units

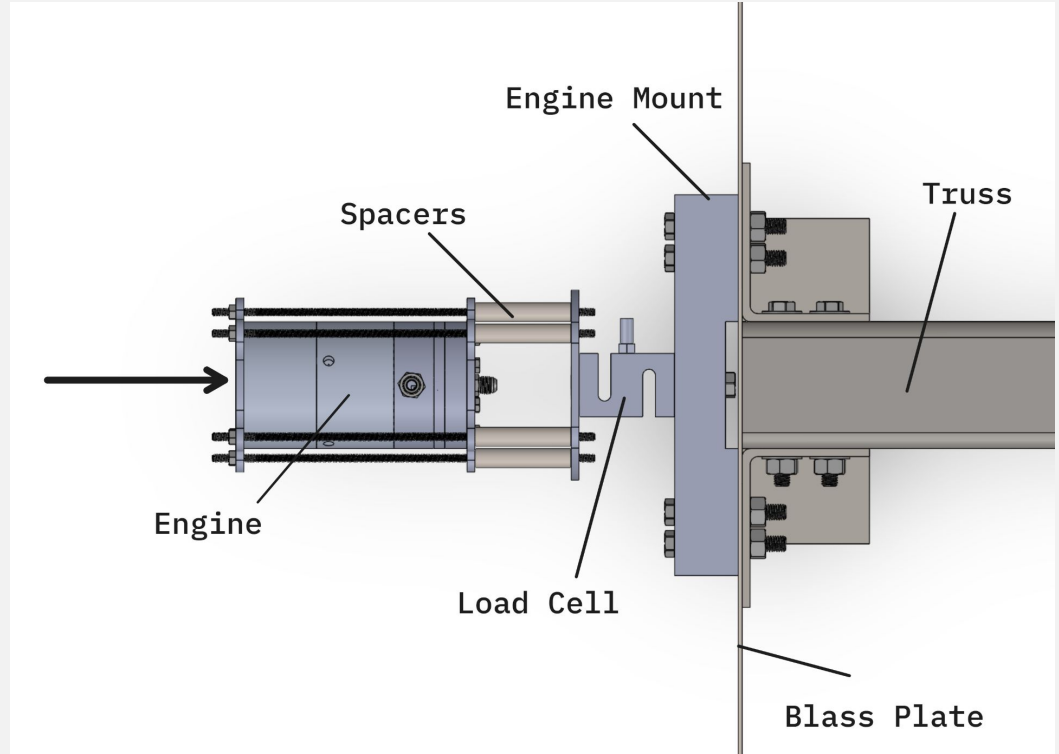
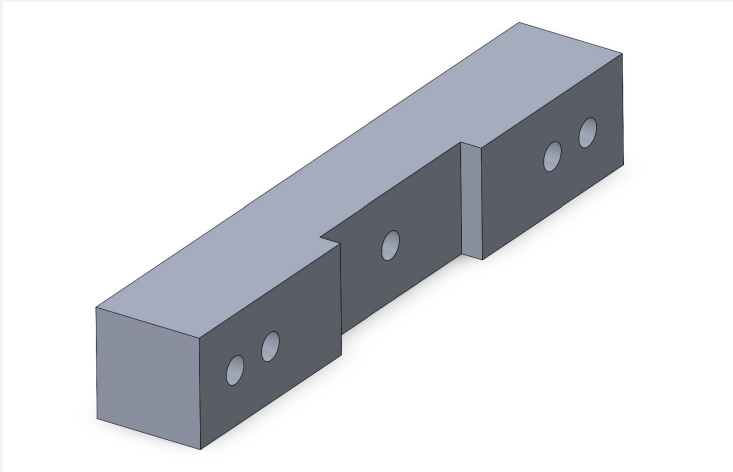
Total Shear Stress (T_tot)	1516.67	psi
Factor of Safety (N)	79.12	
--- Bearing Stress of Gusset Plate ---		
Output	Value	Units

Bearing Stress (bearStress)	77.52	psi
Factor of Safety (N_bear)	464.39	
--- Beam Analysis ---		
Output	Value	Units

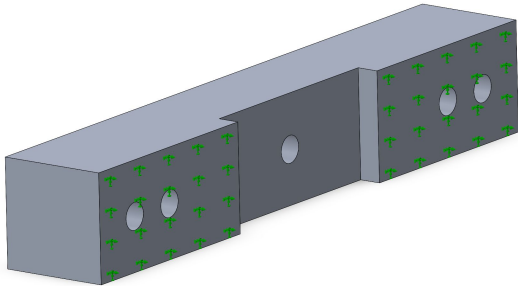
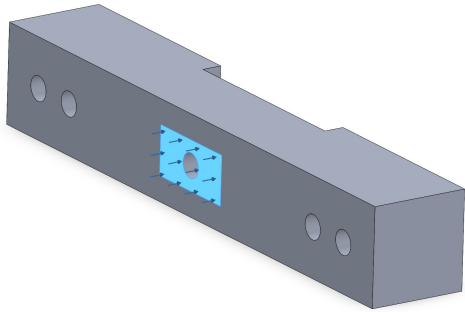
Bending Stress (bendStr)	-210.14	psi
Axial Stress (axStr)	-2.73	psi
Effective Stress (EffStr)	212.87	psi
Factor of Safety (Nbeam)	169.11	
--- Beam Tearout ---		
Output	Value	Units

Tearout Stress (tearStr)	38.76	psi
Effective Tearout Stress (tearEff)	67.14	psi
Factor of Safety (Ntear)	536.23	

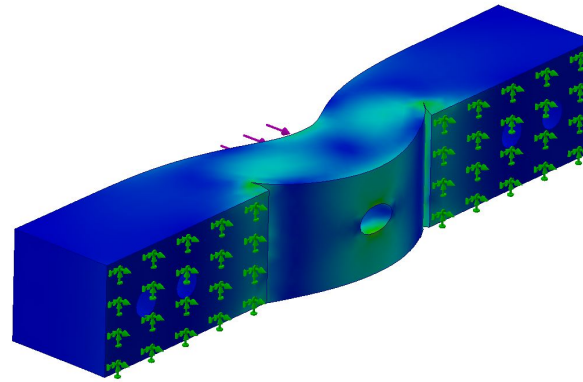
Test Stand - Engine Mounting



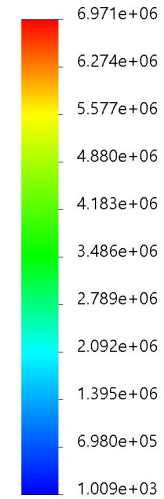
Test Stand - Engine Mounting Cont.1



Model name: Engine mount adapter
Study name: Static 1(-Default-)
Plot type: Static nodal stress Stress1
Deformation scale: 7,935.84



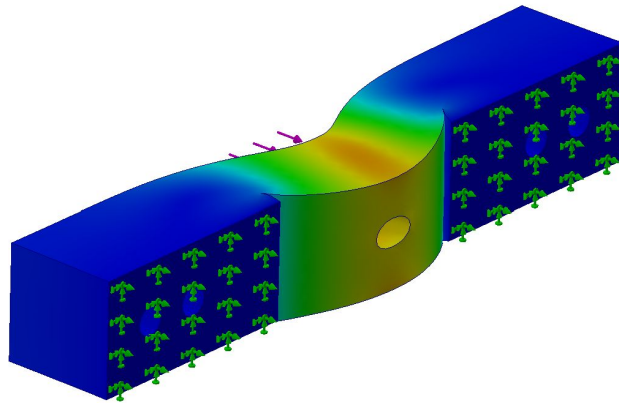
von Mises (N/m²)



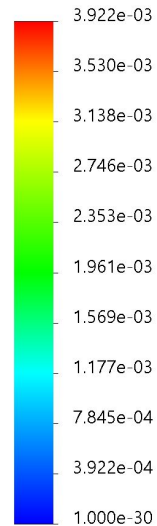
→ Yield strength: 2.750e+08

Test Stand - Engine Mounting Cont.2

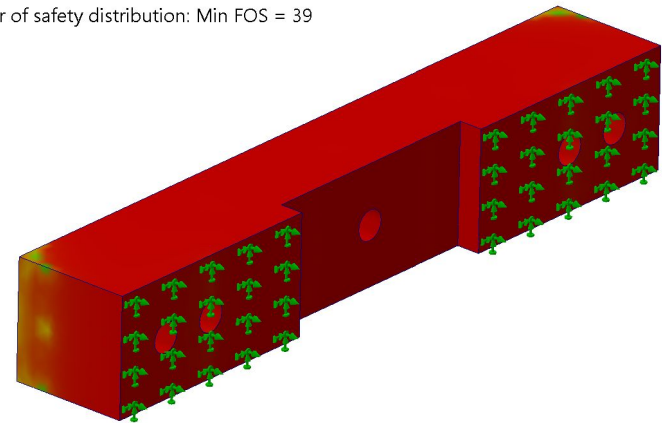
Model name: Engine mount adapter
Study name: Static 1(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 7,935.84



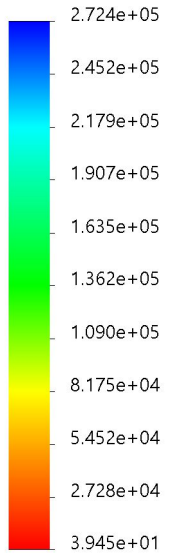
URES (mm)



Model name: Engine mount adapter
Study name: Static 1(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion : Automatic
Factor of safety distribution: Min FOS = 39

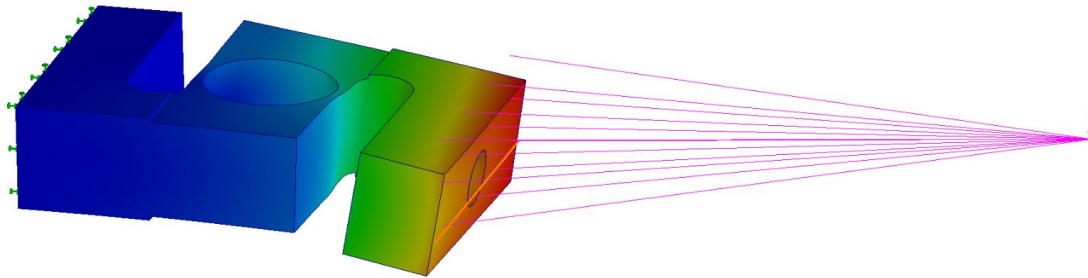


FOS

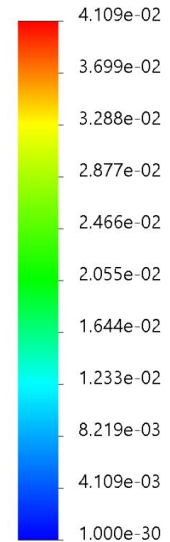


Test Stand - Engine Mounting Cont.3

Model name: loadcellsim
Study name: Static 2(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 189.476



URES (mm)



Test Stand - Ground Fixture

The team is currently talking with KXR and the Kissimmee airport about how to secure the test stand to the ground. The most likely method will consist of drilling tie downs into the ground. More details will be presented at TRR.

Safety

Safety - Summary

Risk Analysis		Severity				
		Negligible (A)	Minor (B)	Moderate (C)	Major (D)	Catastrophic (E)
Probability	Improbable (1)	A1	B1	C1	D1	E1
	Remote (2)	A2	B2	C2	D2	E2
	Occasional (3)	A3	B3	C3	D3	E3
	Likely (4)	A4	B4	C4	D4	E4
	Frequent (5)	A5	B5	C5	D5	E5

Safety - Personnel

ID	Hazard	Cause	Outcome	Before	Mitigation	Verification	After
PH.1	Team member is cut from in-house parts	Sharp edges from manufacturing	Lacerations to skin	C3	All manufactured parts will be chamfered to avoid sharp edges	Manufacturing lead and Chief Engineer make sure that edges are chamfered	A1
PH. 2	Low maneuverability in test prep	Maneuvering around test stand during assembly	Blunt Force Trauma	B4	PPE will be worn during assembly	No personnel allowed in test prep area without PPE	A3
PH. 3	Delayed response time	Miscommunication	Increase hazards covered by EAP and slow emergency response	C4	Standardize communication during test prep and operations	Make sure everyone in operation is trained on standardized communication	B2

Safety - Injector

ID	Hazard	Cause	Outcome	Before	Mitigation	Verification	After
IN.1	Wrong mixture ratio	Orifice clogging, valve timing error, cavitation	Incomplete combustion, soot deposition, or ox-rich combustion erosion	B4	Calibrate inlet pressure for both lines	Verify data during cold flow and water flow tests	B2
IN.2	Erosion of Injector	High heat and localized temperature	Change in the length of the injector element	B5	Short burn of engine	Only provide enough fuel for 3 seconds of burn time	A4
IN.3	Unstable mixing	Only 1 inlet being used. The other one is used for obtaining data	Holes closer to the hose will have higher individual mass flow rates than the rest	B4	Will not be mitigated unless significant effects are observed during water and cold flow tests	Observation on tests	B4

Safety - Nozzle

ID	Hazard	Cause	Outcome	Before	Mitigation	Verification	After
NO.1	Erosion of the nozzle	High Heat and localized temperature	decrease in diameter of the nozzle, decrease on performance.	B5	Short burn of engine	Only provide enough fuel for 3 seconds of burn time	A4
NO.2	Nozzle is blown out of the engine	Threaded rods and plate fail to contain the nozzle fixed to the chamber	Nozzle is blown out	E2	Calculate forces acting on nozzle in the axial direction	FEA analysis	E1

Safety - Test Stand

ID	Hazard	Cause	Outcome	Before	Mitigation	Verification	After
TS.1	Test Stand moves during test	Poor fixing	Test becomes dynamic, poor thrust data gathering	E2	Hand calculation analysis	Confirm with testing location	E1
TS.2	Load Cell bends on engine weight	Engine is too heavy, no calculations on deflection	Thrust vectoring, failure of load cell, possible dynamic test	E1	FEA Analysis	Review and approve FEA results	C1
TS.3	Bolts coming loose due to vibrations	Improper application of locktight/missing locknuts	Test bolts become loose, not proper connection between structural joints	E2	Visual inspection of test stand/following torque specs during assembly	Torque spec checklist	E1

Safety - Feed System

ID	Hazard	Cause	Outcome	Before	Mitigation	Verification	After
FD.1	Overpressurization of the N2O side	Failure of active and passive venting	Go beyond part operating limit	E1	Add relief Valves	Confirm relief valve location and proper set up	C1
FD.2	Overpressurization of the ethanol side	Failure of active vent	Go beyond part operating limit	E1	Add relief Valves	Confirm relief valve location and proper set up	C1
FD.3	Low pressure inlet on air pneumatic valve	Tank was not filled or low pressure remaining from previous test	Failure to open pneumatic valves	B2	Fill the portable tank before test	Use of checklist to verify step	B1
FD.4	N2O check valve fails	Failure of check valve	N2O leaks into the fuel side	D2	Check valves will be tested	Pressure data before and after check valve	D1
FD.5	Incorrect Regulator settings	Operator error during setup or regulator drift	Over/under-pressurization of feed system, off-nominal propellant flow	C2	Define and document target regulator set points	Use of checklist to confirm regulator settings	C1
FD.6	Leaking on fitting connections	Improper assembly, damaged seals, or over/under-torqued fittings	Propellant release, fire hazard, loss of system pressure	D2	Leak check all connections prior to test using pressurized inert gas	Confirm zero pressure drop over hold period	C1

Cleaning Procedure

Cleaning Procedures

Item	Quantity x Size
Liquinox	1 x Quart
Deionized Water	20 Gallons
Air Compressor	1x
Lint Free Paper Towel	1 Roll
Corrosion Inhibitor Bags	50x
Ultrasonic Cleaner	1x
Air Compressor Hose and Fittings	1x
PPE	2x

	<u>Preparation</u>
1	Mix Liquinox solution beforehand with manufacturer recommendations at 1%. So around 2.5 tbsp per gallon.
2	Rinse the bath with DI water to remove possible debris.
3	Inspect parts to make sure larger debris like Teflon residue is removed.
	<u>Cleaning Procedure</u>
1	Place all items into the basket.
2	(Steel) Heat up to 80 degrees Celsius and start the timer to clean for 30 minutes. (Brass) 50 degrees Celsius for 20 minutes.
3	Afterwards turn the cleaner off and take the entire basket and dump it into a bath of just DI water.
4	Agitate the water by moving the basket around.
5	Lay out each piece onto lint free towels.
6	Use an air compressor to blast residue and water off.
7	Both parties inspect each part so there's no leftover residue of detergent, water, or oxidized metal.
8	Items are then placed into VCI bags for storage.

9	The bag is then tagged and then documented within excel.
10	Bags are then placed into a sealed plastic container for storage.
11	The cleaner is set to rest for 20-30 minutes before the next batch. While it is cooling the old solution is drained and then the bath is rinsed with DI water.

Water Flow SOP

Water Flow SOP

Water Flow Procedure

Pre-Fill Check

Ensure all manual ball valves are closed via handle being perpendicular to the flow.

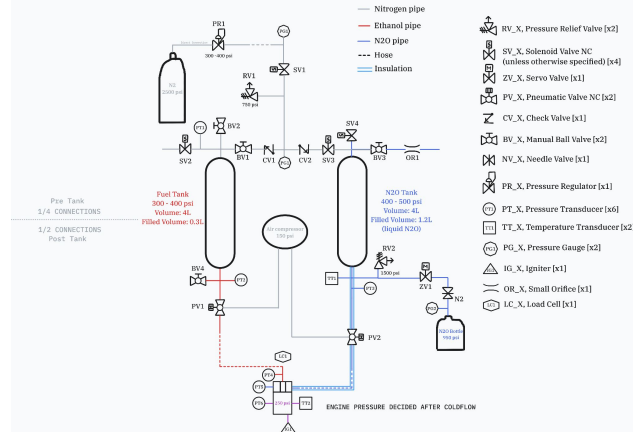
Ensure Pressure Regulators are closed.

Attach air compressor tubes to PV1 and PV2

Turn on the Air Compressor. Let it run until it shuts itself off.

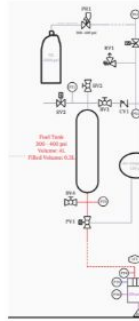
Ground Station turns on the system and ensures full control of all electronically actuated equipment (Solenoids and Servos).

Turn on each individual solenoid and servo and turn them off.



Water Flow SOP cont. Fuel-Side

Fuel Side Test



Fill

Connect fill tube to BV4

Open BV4

Open BV2

Place a bucket of Deionized Water on a ladder and use siphoning to fill the fuel tank.

Close BV4 once tank is completely fill

Close BV2

Disconnect the tube.

Pressurizing Fuel Side

Slowly open PR1 to allow 300-400 psi of Nitrogen through.

Do one last check to make sure all manual valves are closed

Open BV1

Test Stand Personnel are ordered to move away from the test stand.

Chief Engineer gives go for ALL CLEAR.

Open SV1

Let system sit for a minute to read DATA for leaks

(If there are leaks close SV1. Open SV2 to vent out pressure. Personnel return to tighten connections.)

Ground Station gives the clear for testing to go ahead

The Chief Engineer assumes full control of the system.

Test

Open PV1

After All fluids have been dispelled through the injector

Close SV1

Close PV1

Open SV2 to vent out any remaining pressure left in the system

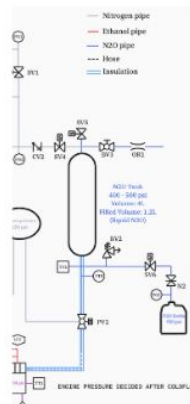
Close SV2 once pressure readings drop to Zero

Personnel are allowed near the test stand.

ENSURE EVERYTHING IS CLOSED BEFORE PROCEEDING TO THE NEXT TEST

Water Flow SOP cont. Oxidizer Side

Oxidizer Side Test



Ensure all manual ball valves are closed via handle being perpendicular to the flow.

Fill

Connect fill tube to ZV1

Open BV3

Open ZV1

Place a bucket of Deionized Water on a ladder and use siphoning to fill the fuel tank.

Close ZV1 once the tank is filled.

Close BV3

Disconnect the fill tube.

Pressurizing

Slowly Open PR1 to allow 400-500 psi of Nitrogen through

Do one last check to make sure all manual valves are closed.

Test Stand personnel are ordered away from the test stand.

Chief Engineer gives go for "ALL CLEAR"

Open SV1

Let system sit for a minute to read DATA for leaks

(If there are leaks close SV1. Open SV5 to vent out pressure. Personnel return to tighten connections.)

Ground Station gives the clear for testing to go ahead

Chief Engineer assumes full control of the system

Test

Open PV2

After All fluids have been dispelled through the injector

Close SV1

Close PV2

Open SV5 to vent out any remaining pressure

Once pressure returns to Zero

Close SVS

Logistics

Logistics - Requirements

Req. ID	Requirement	Rationale	Verification Method	Child Req	Parent Req	Status
LOG-1	The team shall have insurance to perform the test.	Provide financial protection to the club and the testing facility.	Inspection		GEN-4	In Progress
LOG-2	The team shall have permission to test at the facility.	Avoid legal issues.	Inspection		GEN-4	In Progress
LOG-3	The team shall be compliant with propellant transportation regulations.	Ensure safety of personnel.	Inspection		GEN-4	In Progress

Logistics - Insurance

Thank you KXR!

INTERCLUB PARTNERSHIP AGREEMENT

Between

Knights Experimental Rocketry, Inc.

and

University of South Florida Society of Aeronautics and Rocketry

Contract No. KXR-USF-001

Fiscal Year 2026-2027

Effective: July 1, 2026

RECITALS

This Interclub Partnership Agreement (the "**Agreement**"), Contract No. KXR-USF-001, is entered into as of July 1, 2026, by and between **Knights Experimental Rocketry, Inc.**, a Florida nonprofit corporation and 509(a)(2) public charity organized under the laws of the State of Florida, whose principal mailing address is 12760 Pegasus Drive, Building 40, Room 307, Orlando, Florida 32816 ("**KXR**"), and the **University of South Florida Society of Aeronautics and Rocketry**, a registered student organization of the University of South Florida ("**USF SOAR**"), and together with KXR, the "**Parties**").

WHEREAS, KXR is a student-led aerospace engineering organization operating under the authority of the University of Central Florida and holds a commercial general liability insurance policy (Kinsale Insurance Company, Policy No. 0100352560 and its successor policy, the "**Policy**") covering research and development of rocket propulsion systems at KXR's designated testing facility;

WHEREAS, USF SOAR is a student-led rocketry and aeronautics organization operating under the authority of the University of South Florida, engaged in the design, development, and testing of rocket propulsion systems;

WHEREAS, the Parties recognize a mutual interest in collaborative propulsion research and static fire testing, and desire to formalize the terms under which USF SOAR may conduct testing activities at KXR's designated facility;

WHEREAS, KXR is willing to extend access to its designated testing facility and institutional resources to USF SOAR, subject to the terms and conditions set forth herein;

NOW, THEREFORE, in consideration of the mutual covenants and agreements contained herein, and for other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the Parties agree as follows:

1. DEFINITIONS

Thank you!
Questions?